Cleaning and Analyzing Urban Crime Data:

District, Temporal, and Arrest Patterns in an Instructional Incident Dataset, 2020–2024

IE6400 Foundations for Data Analytics Engineering

Project 1

Fall Semester 2026

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Submission Date: October 15, 2026

Abstract

This report applies a complete data analytics workflow to IE6400_Project1_UrbanCrime_Instructional.csv, a fixed instructional dataset of 28,100 urban crime incident records from 2020 through 2024 with 11 fields. Three research questions guided the work: whether crime types are related to district, how recorded incidents changed over the five years, and how the percentage of incidents involving an arrest differs across crime types. We first documented seven data-quality problems, including mixed Boolean labels, inconsistent category labels, exact and hidden duplicates, placeholder dates, and sentinel coordinate values. Cleaning reduced the dataset to 28,000 unique incidents. Methods included cross-tabulated proportions, chi-square tests, a permutation test, days-in-month normalization, and 95% confidence intervals, with key results cross-checked through independent aggregations. The crime-type mix was almost identical across all 12 districts (chi-square = 93.7, df = 99, p = 0.63; Cramér's V = 0.019), and no district deviated from the overall mix by more than random shuffling would produce. Annual totals were stable, ranging from 5,549 to 5,623, even though individual months varied from 371 to 520 incidents with no consistent seasonal peak. Arrest percentages differed strongly by crime type (Cramér's V = 0.325). Narcotics incidents had a 62.0% arrest percentage, about 4.5 times the overall 13.8%, and narcotics, assault, and battery held the top three positions in every year. Because the data are instructional and lack population denominators, all findings are described as count-based patterns in the course dataset, not as crime rates in a real city.

Keywords: *data cleaning, exploratory data analysis, urban crime incidents, chi-square test, permutation test, arrest percentage*

1. Introduction and Research Questions

1.1 Motivation and Scope

Police incident records are a common source of information about urban crime, but they are rarely ready for analysis when first exported. Labels are typed inconsistently, incidents are entered more than once, and dates or coordinates are filled with placeholders when the true value is unknown. If these problems are missed, simple summaries such as "the most common crime type" can be wrong without anyone noticing. This project therefore treats cleaning and validation as part of the analysis itself.

The scope of the project is limited to the dataset supplied for the course, IE6400_Project1_UrbanCrime_Instructional.csv. The file was designed for data-cleaning practice and is modeled on the structure of the City of Chicago's public crime data (City of Chicago, n.d.), but it is not a copy of official police records. For that reason, every result in this report is a statement about patterns in the course dataset, and none of them should be read as a claim about crime in Chicago or any other real city.

1.2 Analytical Objective

Our objective was to understand how recorded incidents in the dataset relate to three dimensions: geography (police district), time (year and month), and outcome (whether an arrest was recorded). Specifically, we wanted to know whether certain crime types are concentrated in particular districts, whether the volume of recorded incidents changed between 2020 and 2024, and whether some crime types are much more likely than others to involve an arrest. Answering these questions required first producing a clean, documented version of the data in which each incident is counted exactly once and each category is labeled consistently.

1.3 Research Questions

RQ1. Is there a relationship between crime types and the districts in which they occur?

RQ2. How has the number of recorded crime incidents changed from 2020 through 2024?

RQ3. How does the percentage of incidents involving an arrest differ across crime types?

1.4 Analysis Plan

RQ1 is a categorical comparison between primary_type and district. Because districts recorded different numbers of incidents, we compared the share of each crime type within each district rather than raw counts. RQ2 is a temporal analysis that uses incident_datetime aggregated to yearly and monthly totals, with an adjustment for the different number of days in each month. RQ3 combines primary_type with the arrest indicator, first pooled across all five years and then year by year to check whether the pattern is stable. We did not integrate any external data. None of our research questions required population figures, and we deliberately avoided the word "rate" because the dataset contains only incident counts (see Section 7).

2. Dataset and Data Quality

2.1 Source and Provenance

The dataset was provided through the course website by the instructor, Prof. Sivarit Sultornsanee, for IE6400 Foundations for Data Analytics Engineering in Fall 2026. It was loaded directly from the supplied CSV with pandas (McKinney, 2010). The loaded table contained 28,100 rows and 11 columns, which matches the size stated in the assignment. The City of Chicago Data Portal dataset Crimes – 2001 to Present (dataset ID ijzp-q8t2) was used only as a reference for what each field means. No data were downloaded from the portal, and the course CSV was not replaced with a live download. The raw file was never modified: all cleaning was done on a separate copy (df_clean), so the original table (df_raw) was always

available for before-and-after comparisons. The cleaned table was exported as IE6400_Project1_Cleaned.csv.

2.2 Variables and Analytical Roles

Each row represents one recorded incident. Table 1 lists the fields and how we treated each one. An important early decision was that incident_id, district, and community_area are stored as numbers but are not measurements. District 12 is not "greater than" District 5 in any meaningful sense, so averages or other arithmetic on these columns would have no interpretation. We treated incident_id as an identifier and district and community_area as categorical geographic codes. The x_coordinate and y_coordinate fields are the only true spatial measurements, and arrest and domestic are binary indicators.

Table 1. Dataset fields and their analytical roles

Field	Role	Description
incident_id	Identifier	Unique number assigned to each incident
incident_datetime	Temporal	Date and time of the incident
primary_type	Categorical	General crime category (10 categories after cleaning)
description	Categorical	More specific description of the offense
location_description	Categorical	Type of location (10 categories after cleaning)
arrest	Binary indicator	Whether an arrest was recorded
domestic	Binary indicator	Whether the incident was flagged as domestic-related
district	Geographic code	Police district (12 districts)
community_area	Geographic code	Community area (30 areas)
x_coordinate	Spatial measurement	Horizontal projected coordinate
y_coordinate	Spatial measurement	Vertical projected coordinate

2.3 Initial Quality Assessment

Before cleaning, the parseable dates ran from January 1, 2020 to December 31, 2024. Theft was the largest category (6,707 records under the uppercase label), followed by battery (4,717) and criminal damage (3,958), and street was the most common location (6,227). Each district held between 1,964 and 2,770 raw records, each community area between 872 and 995, and about 14% of records involved an arrest and 10% were domestic-related. Inspection identified seven quality problems.

Finding 1: inconsistent Boolean values. The arrest and domestic columns each contained four labels instead of two. Arrest held False (24,164), True (3,856), No (61), and Yes (19), and domestic held False (25,182), True (2,838), No (74), and Yes (6). The values meant the same thing but would be counted as separate groups.

Finding 2: inconsistent crime categories. primary_type contained 20 labels for what were actually 10 categories, because some records were entered in lowercase (for example, "theft" appeared 40 times and "battery" 25 times alongside their uppercase versions).

Finding 3: exact duplicate rows. The data contained 99 rows that were identical in every column to another row, including the incident_id.

Finding 4: a hidden duplicate. incident_id was repeated 100 times, one more than the number of exact duplicates. The extra case was a pair of records for incident 10022388 that matched in every field except domestic, where one record said False and the other said No. This showed that the Boolean inconsistency in Finding 1 was hiding a duplicate and that the order of cleaning steps mattered.

Finding 5: placeholder dates. No values in incident_datetime were technically null, but 35 records contained the text "not recorded" instead of a date.

Finding 6: invalid and missing coordinates. 784 records were missing x_coordinate and 788 were missing y_coordinate. In 20 records both were missing, 764 were missing only X, and 768 were missing only Y. In addition, 25 records had an X value of 0 and 25 had a Y value of −999. These sentinel values lay far outside the range of the other coordinates and pulled the minimum of each column to an impossible value.

Finding 7: inconsistent location labels and missing location context. location_description contained 20 non-missing labels for 10 locations, because some values had leading or trailing spaces (for example, " STREET " appeared 30 times). The column also had 503 missing values. district was missing in 168 records and community_area in 336.

2.4 Important Limitations of the Dataset

Three properties of the dataset limit what can be concluded from it: it is instructional rather than official, it contains counts with no population or area denominators, and the arrest field records only whether an arrest was linked to an incident. These limitations are discussed in Section 7.

3. Methodology

3.1 Workflow and Reproducibility

All work was done in a single Jupyter Notebook using pandas, NumPy (Harris et al., 2020), Matplotlib (Hunter, 2007), and SciPy (Virtanen et al., 2020). The notebook follows the order of the assignment tasks: loading, inspection, cleaning, exploratory analysis, and interpretation. Every cleaning step prints the affected counts before and after the change. The only random procedure, the permutation test in RQ1, uses a fixed seed (42), so the notebook produces the same output each time it is run.

3.2 Cleaning and Transformation

Cleaning was applied in a deliberate order. First, "Yes" and "No" were replaced with "True" and "False" in both indicator columns, and the columns were converted to the Boolean type. Second, primary_type was converted to uppercase, and leading and trailing spaces were removed from location_description. Third, incident_datetime was parsed with pd.to_datetime using errors="coerce", which turned the 35 "not recorded" strings into missing timestamps (NaT) instead of guessing a date. From the parsed timestamp we derived year, month, day of week, and hour. Fourth, the coordinate sentinels (X = 0 and Y = −999) were replaced with missing values. After this step the coordinates fell within a plausible range (X from 1,134,633 to 1,226,535 and Y from 1,815,182 to 1,908,283).

Duplicates were removed last, and this ordering was intentional. If duplicates had been dropped before the Boolean values were standardized, the hidden duplicate in Finding 4 would have survived, and incident 10022388 would have been counted twice. After standardization, a full-row duplicate check found 100 duplicates, which matched the 100 repeated incident_id values exactly. Removing them reduced the data from 28,100 to 28,000 rows, and a repeated check found no remaining duplicates by either method. We used the full row together with incident_id as the duplicate key. Records with different incident_id values were treated as separate incidents, even when they shared a date, category, or location, because two real incidents can easily have similar attributes. The description field contained 21 consistent labels and did not need recoding.

3.3 Missing-Value Strategy

No records were deleted because of missing values, and no values were imputed. Imputing a district or community area would assign incidents to areas that the data cannot confirm, and the mean or median of a

categorical code is meaningless. Imputing dates or coordinates would create information that does not exist. Instead, each analysis excluded only the records missing a field that analysis required. RQ1 used the 27,832 incidents with a known district (168 excluded). The time-based parts of RQ2 and RQ3 used the 27,965 incidents with a valid date (35 excluded). The pooled arrest analysis in RQ3 used all 28,000 incidents, because both primary_type and arrest were complete. The consequence of this approach is that each analysis runs on a slightly different denominator, so we report the denominator alongside each result.

3.4 Encoding and Scaling

We did not apply normalization, standardization, or one-hot encoding, because counts, proportions, and contingency tables work directly with the original categories. These transformations would only be needed for a model, and we did not build one. district, community_area, year, month, and hour were stored as nullable integers so they display as whole numbers while still allowing missing values.

3.5 Analytical Methods

For RQ1 we built a crime-type by district cross-tabulation and converted each district's column to percentages, so every district's mix sums to 100% regardless of its size. Each district's mix was compared with the overall mix. We then tested for an association in two ways. The first was a chi-square test of independence with Cramér's V as a measure of effect size (Agresti, 2019; Cramér, 1946). The second was a permutation test that does not rely on the chi-square approximation (Good, 2005). For each district we recorded the largest absolute difference, in percentage points, between its crime-type shares and the overall shares, and compared it with the 95th percentile of the same statistic across 1,000 random shuffles of the district labels (which keep each district's total fixed).

For RQ2, incidents were counted by year with year-over-year percentage changes, and cross-tabulated by month and year. Monthly counts were also divided by the days in each month to give an average daily count, with February 2020 and 2024 treated as 29-day months.

For RQ3, the arrest percentage for each crime type was the share of its incidents with arrest = True. A 95% confidence interval was attached to each percentage using the normal approximation for a proportion. A chi-square test with Cramér's V was used to measure the association between crime type and arrest. Arrest percentages were then recalculated separately for each year from 2020 to 2024 and ranked within each year to check whether the pooled pattern held over time.

3.6 Validation Approach

Validation had two parts. First, we confirmed row counts, category counts, date ranges, and missing-value totals after cleaning. Second, we cross-checked important results with an independent calculation, such as totals from a different grouping or a resampling test alongside a parametric one. Both are reported in Section 5.

4. Results and Exploratory Analysis

4.1 Data-Quality Outcomes

Table 2 summarizes each problem found in Section 2.3, the action taken, and the result. Table 3 compares missing values before and after cleaning. Some missing-value counts went up after cleaning. This is expected and correct: placeholder text and sentinel numbers were converted into explicit missing values so they could no longer be mistaken for real data. The small decreases between the intermediate and final counts (for example, location_description going from 503 to 501) occurred because a few of the removed duplicate rows also had missing values.

Table 2. Before-and-after data-quality summary

Issue	Before cleaning	Action taken	After cleaning
Total rows	28,100	Removed duplicate copies	28,000
Mixed Boolean labels	4 labels each in arrest and domestic	Mapped Yes/No to True/False; converted to Boolean	2 values each
primary_type labels	20 (case variants)	Converted to uppercase	10
location_description labels	20 non-missing (whitespace)	Stripped leading/trailing spaces	10
Exact duplicates	99 (plus 1 hidden)	Standardized Booleans, then drop_duplicates()	0
Repeated incident_id	100	Removed with duplicates	0
Placeholder dates	35 "not recorded"	Parsed with errors="coerce" to NaT	35 missing, retained
Coordinate sentinels	25 X = 0; 25 Y = −999	Replaced with missing values	0 sentinels remaining
Columns	11	Added year, month, day_of_week, hour	15

Table 3. Missing values by column before and after cleaning

Column	Raw data	Cleaned data	Reason for change
incident_datetime	0 (35 placeholders)	35	Placeholder text converted to NaT
location_description	503	501	Two missing values were in removed duplicates
district	168	168	No change
community_area	336	336	No change
x_coordinate	784	805	+25 sentinels, minus missing values in duplicates
y_coordinate	788	807	+25 sentinels, minus missing values in duplicates

4.2 RQ1: Crime Types and Districts

Districts recorded different numbers of incidents. District 7 had the most (2,754) and District 5 had the fewest (1,956), so District 7 recorded 41% more incidents (Appendix Table A1). This is a difference in volume, not in crime mix. Theft made up 24.4% of District 7's incidents and 24.9% of District 5's. Without population or area data, the difference in volume also cannot be read as a difference in risk, which is why the rest of this section compares percentages only.

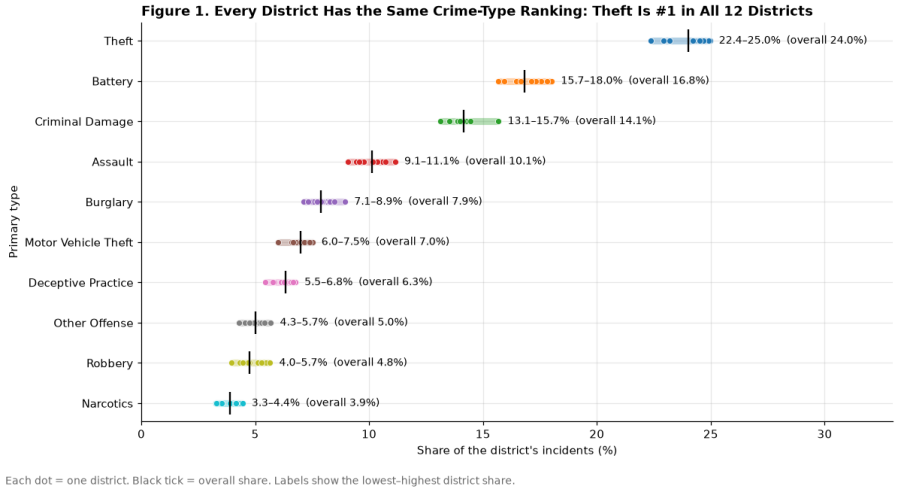


Figure 1. *Share of each district's incidents by crime type. Each dot is one district, the shaded bar spans the lowest to highest district, and the black tick marks the overall share.*

Figure 1 shows that the crime-type ranking is the same in every district. Theft was the most common type in all 12 districts, at between 22.4% and 25.0% of each district's incidents. Its lead over the second-ranked type was never smaller than 4.9 percentage points, so its first-place position was never close. The top four types (theft, battery, criminal damage, and assault) appeared in the same order in all 12 districts. The spread across districts was small for every type: the widest range was 2.6 percentage points (theft and criminal damage), and the narrowest was 1.2 percentage points (narcotics). In practical terms, about one incident in four is theft, one in six is battery, and one in seven is criminal damage in every district.

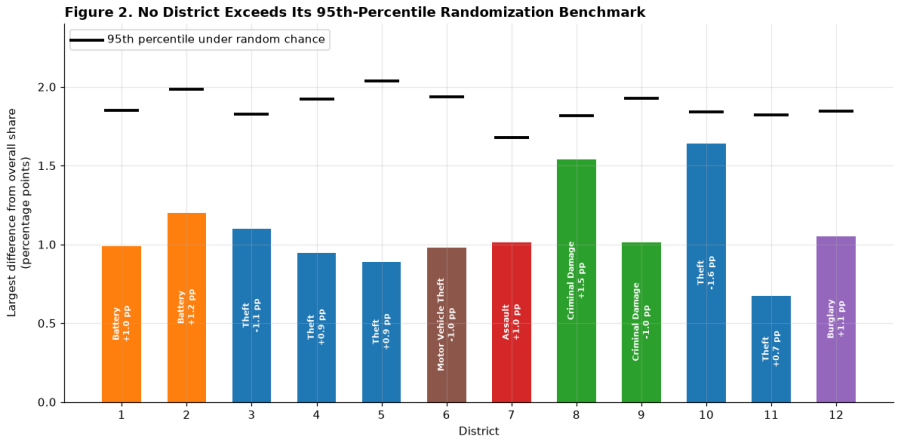


Figure 2. *Largest difference between each district's crime-type mix and the overall mix (bars), compared with the 95th percentile of the same statistic across 1,000 random shuffles of the district labels (black marks).*

Figure 1 shows that the differences are small, but not whether they are larger than chance. Figure 2 addresses this. The largest deviation anywhere in the table was theft in District 10, which was 1.6 percentage points below the overall theft share. In all 12 districts, the largest deviation was below the 95th-percentile benchmark from the shuffled data. In 74% of the shuffles, at least one district showed a deviation as large as 1.6 percentage points, so the observed result is typical of random variation. The chi-square test agreed (chi-square = 93.7, df = 99, p = 0.63), and Cramér's V was 0.019, indicating almost no association. The smallest expected cell count was 76, well above the usual minimum of 5, so the chi-square approximation was appropriate. We therefore found no evidence of a relationship between crime type and district in this dataset. Districts differ in how many incidents they record, but not in what kinds of incidents they record.

4.3 RQ2: Change in Recorded Incidents, 2020–2024

Table 4. Recorded incidents by year (n = 27,965 with a valid date)

Year	Incidents	Change from previous year
2020	5,623	—
2021	5,549	−1.32%
2022	5,572	+0.41%
2023	5,611	+0.70%
2024	5,610	−0.02%

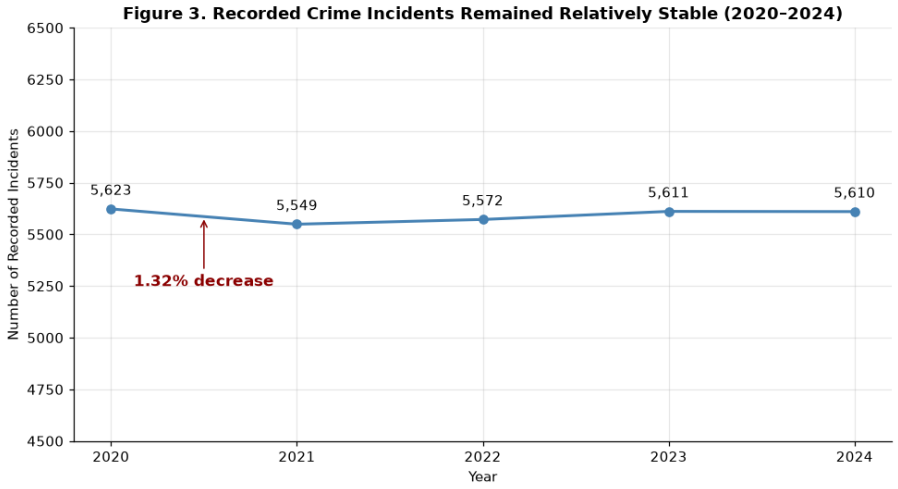


Figure 3. *Recorded incidents per year, 2020–2024. The vertical axis starts at 4,500 to make the year-to-year differences visible.*

Annual totals were very stable (Table 4; Figure 3). The largest year-over-year change was a 1.32% decrease from 2020 to 2021, from 5,623 to 5,549 incidents. Totals then rose slightly in 2022 and 2023 and were essentially unchanged in 2024. Over the full period, 2024 recorded 13 fewer incidents than 2020, a decrease of about 0.23%. A difference of this size is smaller than the number of records excluded for missing dates (35), so the data do not support describing the period as either increasing or decreasing.

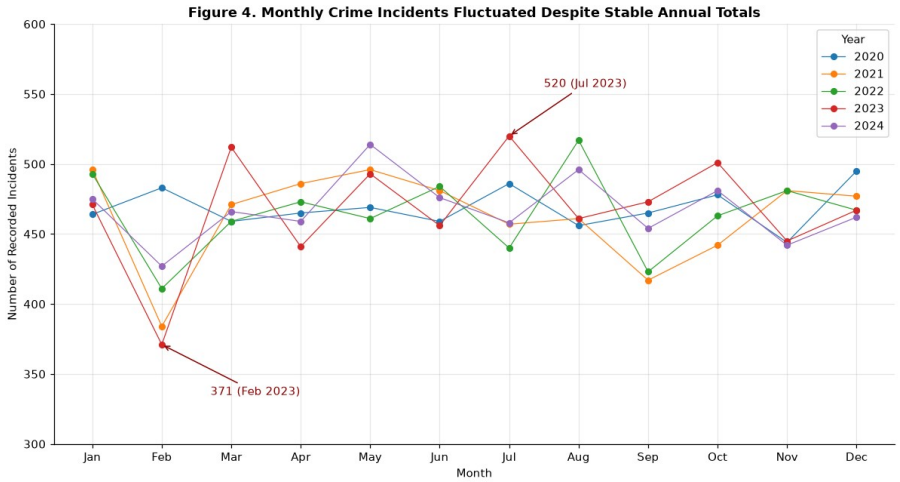


Figure 4. *Monthly recorded incidents by year. Labels mark the lowest month (February 2023) and highest month (July 2023) in the five-year period.*

Stable annual totals can hide variation within a year, so Figure 4 breaks the counts down by month (full counts in Appendix Table A2). Monthly counts ranged from 371 (February 2023) to 520 (July 2023), a gap of 149 incidents within a single year. February had the lowest raw count in four of the five years, with 2020 as the exception. However, February is also the shortest month. After dividing by the number of days (Appendix Table A3), February 2023 was still the lowest month in the dataset at 13.25 incidents per day, but February 2020 was among the highest at 16.66 per day. The same calendar month therefore appears near both extremes, which argues against a February effect. The month with the most incidents also changed every year: December in 2020, January and May (tied) in 2021, August in 2022, July in 2023, and May in 2024. We therefore found no consistent seasonal peak in the dataset.

We treated February 2023 and July 2023 as the main anomalies. Both values appear in two independent aggregations (the month-by-year table and the daily-average table), and the month-length adjustment confirmed that February 2023 is low on a daily basis, not just because February is short. The dataset contains no variables that could explain these months, so we report them as unusual values without assigning a cause.

4.4 RQ3: Arrest Percentages by Crime Type

Table 5. Arrest percentage by crime type with 95% confidence intervals (n = 28,000)

Crime type	Incidents	Arrests	Arrest %	95% CI
Narcotics	1,092	677	62.0	59.1–64.9
Assault	2,832	655	23.1	21.6–24.7
Battery	4,720	869	18.4	17.3–19.5
Other offense	1,405	199	14.2	12.3–16.0
Robbery	1,331	164	12.3	10.6–14.1
Burglary	2,210	230	10.4	9.1–11.7
Criminal damage	3,966	335	8.4	7.6–9.3
Theft	6,720	496	7.4	6.8–8.0
Motor vehicle theft	1,956	139	7.1	6.0–8.2
Deceptive practice	1,768	95	5.4	4.3–6.4
All incidents	28,000	3,859	13.8	—

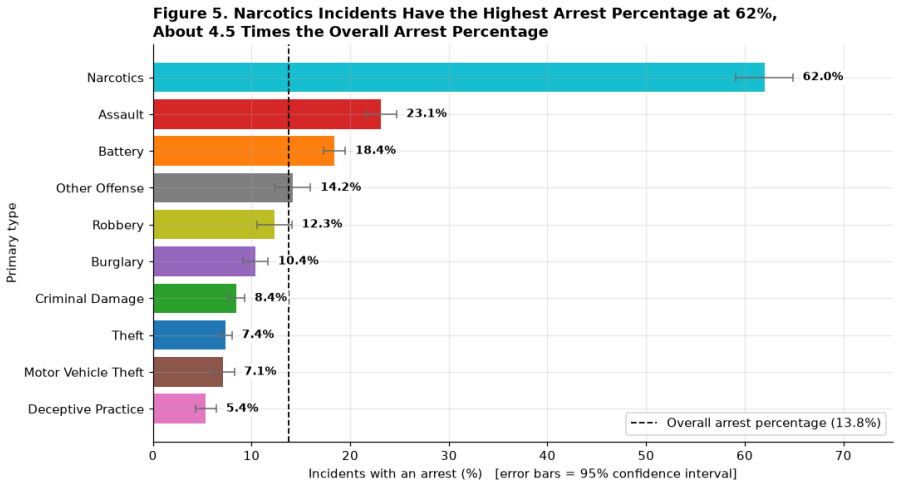


Figure 5. *Percentage of incidents with an arrest by crime type, pooled over 2020–2024. Error bars show 95% confidence intervals; the dashed line marks the overall arrest percentage.*

Arrest percentages varied widely across crime types (Table 5; Figure 5). Narcotics stood far above every other category: 677 of 1,092 narcotics incidents (62.0%) involved an arrest. That is about 4.5 times the overall percentage of 13.8%, about 5.2 times the 11.8% for all other types combined, and about 2.7 times the next-highest type, assault (23.1%). Only four types were above the overall percentage: narcotics, assault, battery (18.4%), and other offense (14.2%), and other offense was only slightly above it. The lowest percentages were for theft (7.4%), motor vehicle theft (7.1%), and deceptive practice (5.4%), a gap of 56.6 percentage points between the highest and lowest types.

The confidence intervals show which differences are clear. The narcotics interval (59.1–64.9%) is far above every other interval, and the assault and battery intervals do not overlap with each other or with any lower category. In contrast, the intervals for criminal damage, theft, and motor vehicle theft overlap, so we do not treat the small differences among these three as meaningful. The chi-square test showed a strong association between crime type and arrest (chi-square = 2,958.6, df = 9, $p < 0.001$). Cramér's V was 0.325, a moderate effect and about 17 times larger than the value for crime type and district in RQ1.

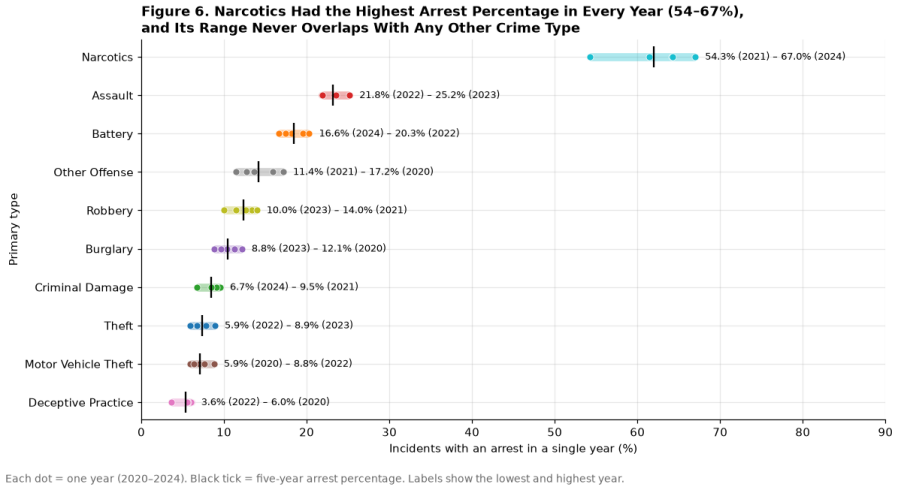


Figure 6. *Arrest percentage by crime type in each year, 2020–2024. Each dot is one year, the shaded bar spans the lowest to highest year, and the black tick marks the five-year percentage.*

Because Figure 5 pools five years, it could hide changes over time. Figure 6 shows each type's yearly values. The narcotics percentage ranged from 54.3% (2021) to 67.0% (2024). Even its lowest year was 29.1 percentage points above the highest single-year value of any other type (assault at 25.2% in 2023). Narcotics, assault, and battery ranked first, second, and third in every year (Appendix Table A4). The lower categories changed position more often. For example, theft ranked anywhere from sixth to ninth, because their percentages are close together and small changes can reorder them. The 2021 dip in narcotics is the largest single-year movement in the table. Narcotics averages only about 220 incidents per year, compared with about 1,340 for theft, so a change of 10 to 20 arrests moves its yearly percentage by several points, and we interpret that dip with caution.

5. Evaluation and Validation

5.1 Post-Cleaning Checks

After cleaning we confirmed that the dataset contained 28,000 rows and 15 columns, that 28,000 is exactly the original 28,100 minus the 100 repeated incident_id values, and that no full-row duplicates or repeated identifiers remained. primary_type and location_description each contained 10 categories, arrest and domestic contained only True and False, and no coordinate sentinels remained. The valid dates still ran from January 1, 2020 to December 31, 2024, so parsing the dates had not shifted or dropped any valid values.

5.2 Independent Cross-Checks

Table 6 lists the cross-checks used to confirm key results; they are implemented in Task 7 of the notebook (notebook Table 8). In every case, two calculations that do not depend on each other produced the same answer.

Table 6. Independent cross-checks of key results

Result checked	Method A	Method B	Outcome
Annual totals (RQ2)	value_counts() on year	Column sums of the month × year cross-tabulation	Both give 5,623 / 5,549 / 5,572 / 5,611 / 5,610
Records with a valid date (RQ2)	Sum of annual totals: 27,965	28,000 rows − 35 missing dates	27,965 in both
District coverage (RQ1)	Column sums of the crime-type × district table: 27,832	Rows missing district: 168	27,832 + 168 = 28,000
No district association (RQ1)	Chi-square test: p = 0.63	Permutation test: all 12 districts below the 95th percentile; 74% of shuffles ≥ observed	Both indicate no association
Overall arrest percentage (RQ3)	Mean of the arrest column: 13.8%	Sum of arrests by type (3,859) ÷ 28,000	13.8% in both
Duplicate removal	Full-row duplicated(): 100 → 0	incident_id duplicated(): 100 → 0	Both keys agree
Arrest ranking stability (RQ3)	Pooled five-year ranking	Separate ranking for each year	Top three identical in all five years

5.3 Effect of Data-Quality Issues on the Findings

Several cleaning decisions directly changed the results. Without standardizing capitalization, every crime type would have been split in two; theft alone would have lost 40 raw records to a lowercase category. Without removing duplicates, 100 incidents would have been counted twice, 16 of them arrests, which would have shifted the RQ3 percentages. Removing duplicates before fixing the Boolean labels would have missed one duplicate entirely.

The date problem is most important for RQ2. The 35 records with placeholder dates are a small share of the data (0.13%), but the entire 2020-to-2024 change was only 13 incidents. If those 35 incidents belong to particular years, they could change the sign of the five-year difference. Our conclusion that annual totals were stable holds either way, but this is why we do not describe the period as a decline. The 168 records with a missing district are 0.6% of the data. They could only change the RQ1 conclusion if they were heavily concentrated in one district and one crime type, and nothing in the data suggests this, so we consider the conclusion robust to this exclusion. Coordinates were not used in our analyses, so their missing values do not affect our results, but they would matter for future mapping work.

6. Discussion

Taken together, the three questions show where variation exists in this dataset and where it does not. Geography explains almost nothing about what type of crime an incident is: districts differ in volume, but their crime-type mixes are statistically indistinguishable. Time also explains little at the annual level, since yearly totals stayed within a band of about 74 incidents. Crime type, however, is strongly associated with the arrest outcome. The contrast between RQ1 and RQ3 is useful because both rely on the same kind of test. A Cramér's V of 0.019 for type and district, compared with 0.325 for type and arrest, shows that the same method can find a real association in one part of the data and rule one out in another.

The narcotics result has a plausible explanation that the data cannot confirm. In real reporting systems, narcotics incidents are often recorded when an officer finds drugs during a stop, so the incident and arrest happen together, while theft and deceptive practice are usually reported by victims later, often with no suspect present. Because the dataset has no information on how incidents were detected, we treat this as an association, not a finding about policing.

The time-series results show why the choice of aggregation matters. A yearly chart alone suggests that nothing happened, while the monthly view shows swings of up to 149 incidents within a year. Normalizing by days in the month changed the interpretation of February from "consistently low" to "inconsistent", which prevented a misleading seasonal claim.

Finally, the hidden duplicate was found only because two independent checks, full-row duplicates and repeated incident_id values, disagreed by one, which shows the value of checking the same property in more than one way.

7. Limitations and Responsible Interpretation

Instructional data. The dataset was built for teaching and is not an official police record. The absence of geographic differences in RQ1, in particular, may reflect how the data were generated rather than anything about real districts. None of the findings should be used to describe crime in Chicago or to compare real neighborhoods.

Counts, not rates. All geographic and temporal comparisons are based on incident counts. Without population, area, or exposure denominators, a district with more incidents cannot be called more dangerous, and a month with more incidents cannot be said to have a higher crime rate. We did not add external data, so we have avoided the term "rate" throughout.

Recorded incidents only. The data describe incidents that were recorded, not all crime that occurred. Differences in reporting by crime type, place, or time cannot be separated from differences in actual offending.

Meaning of the arrest field. The arrest indicator shows only whether an arrest was linked to an incident record. It is not a measure of police effectiveness, clearance, or how often offenders are caught, and the association with crime type is not causal.

Missing and excluded data. Each analysis excluded records missing a required field (168 for district and 35 for date). These exclusions are small, but the RQ2 conclusion is sensitive to them because the five-year change was so small. The 501 records with no location and the roughly 800 records with incomplete coordinates would limit any location or mapping analysis.

Analytical scope. We did not analyze day-of-week or hourly patterns, community areas, location_description, or the domestic indicator in depth, even though the cleaned dataset supports these analyses. The monthly analysis is descriptive, and we did not formally test whether the monthly variation exceeds what would be expected by chance. The confidence intervals in RQ3 use the normal approximation, which is reasonable here because every category has at least 95 arrests and 95 non-arrests.

8. Conclusion

After correcting seven data-quality problems and reducing the dataset from 28,100 records to 28,000 unique incidents, we answered three research questions about the course dataset. First, crime type is not related to district: every district has the same crime-type ranking, and the small differences that exist are consistent with chance ($p = 0.63$; Cramér's $V = 0.019$). Second, the number of recorded incidents was stable from 2020 to 2024, at between 5,549 and 5,623 per year, despite month-to-month swings with no consistent seasonal peak. Third, arrest percentages depend strongly on crime type: narcotics (62.0%) far exceeds every other category, followed by assault (23.1%) and battery (18.4%), and these three held the same positions in every year. These conclusions were supported by independent cross-checks and apply only to the instructional dataset.

Natural next steps would be to analyze day-of-week and hourly patterns, examine arrest and domestic percentages by location type, compare community areas, and test monthly variation formally. Adding justified population data with matching boundaries and years would allow count comparisons to be converted into rates, and a predictive model for the arrest outcome could be evaluated against a majority-class baseline on held-out data.

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Appendix A. Supplementary Tables

Table A1. Incident counts and top-three crime-type shares by district (n = 27,832)

District	Incidents	Theft (%)	Battery (%)	Criminal damage (%)
1	2,274	24.8	17.8	14.4
2	1,970	22.9	18.0	13.9
3	2,488	22.9	17.8	14.3
4	2,264	25.0	16.5	14.2
5	1,956	24.9	16.7	14.3
6	2,228	23.2	15.9	14.5
7	2,754	24.4	15.9	13.5
8	2,474	24.3	15.7	15.7
9	2,201	24.2	17.5	13.1
10	2,499	22.4	17.3	13.9
11	2,265	24.7	17.1	14.1
12	2,459	24.5	15.9	14.0

Table A2. Recorded incidents by month and year

Month	2020	2021	2022	2023	2024
Jan	464	496	493	471	475
Feb	483	384	411	371	427
Mar	459	471	459	512	466
Apr	465	486	473	441	459
May	469	496	461	493	514
Jun	459	481	484	456	476
Jul	486	457	440	520	458
Aug	456	461	517	461	496
Sep	465	417	423	473	454
Oct	478	442	463	501	481
Nov	444	481	481	445	442
Dec	495	477	467	467	462
Total	5,623	5,549	5,572	5,611	5,610

Table A3. Average daily recorded incidents by month and year

Month	2020	2021	2022	2023	2024
Jan	14.97	16.00	15.90	15.19	15.32
Feb	16.66	13.71	14.68	13.25	14.72
Mar	14.81	15.19	14.81	16.52	15.03
Apr	15.50	16.20	15.77	14.70	15.30
May	15.13	16.00	14.87	15.90	16.58

Month	2020	2021	2022	2023	2024
Jun	15.30	16.03	16.13	15.20	15.87
Jul	15.68	14.74	14.19	16.77	14.77
Aug	14.71	14.87	16.68	14.87	16.00
Sep	15.50	13.90	14.10	15.77	15.13
Oct	15.42	14.26	14.94	16.16	15.52
Nov	14.80	16.03	16.03	14.83	14.73
Dec	15.97	15.39	15.06	15.06	14.90

Table A4. Arrest percentage by crime type and year (rank within year in parentheses)

Crime type	2020	2021	2022	2023	2024
Narcotics	64.2 (1)	54.3 (1)	64.2 (1)	61.4 (1)	67.0 (1)
Assault	23.4 (2)	23.5 (2)	21.8 (2)	25.2 (2)	21.9 (2)
Battery	19.5 (3)	18.2 (3)	20.3 (3)	17.4 (3)	16.6 (3)
Other offense	17.2 (4)	11.4 (5)	13.6 (4)	12.7 (4)	15.9 (4)
Robbery	13.4 (5)	14.0 (4)	12.7 (5)	10.0 (5)	11.5 (5)
Burglary	12.1 (6)	11.3 (6)	10.4 (6)	8.8 (7)	9.7 (6)
Criminal damage	8.5 (7)	9.5 (7)	9.1 (7)	8.5 (8)	6.7 (8)
Theft	7.6 (8)	6.7 (9)	5.9 (9)	8.9 (6)	7.8 (7)
Motor vehicle theft	5.9 (10)	7.0 (8)	8.8 (8)	7.7 (9)	6.3 (9)
Deceptive practice	6.0 (9)	6.0 (10)	3.6 (10)	5.7 (10)	5.5 (10)